| 姓名: ( ) 班级: 学号: | FOON YEW HIGH SCHOOL – KULAI<br>MATHEMATICS FINAL EXAMINATION<br>SENIOR MIDDLE THREE SCIENCE<br>QUESTIONS SHEET<br>24/09/2026 8:00am – 10:00am | 成绩:<br><br>家长签名: |
|---|---|---|

## Part 1 – Multiple Choice Questions – Answer ALL questions [10 x 3%]

1. How many solutions does the equation $\sin 5x + \sin x = 0, x \in [0, \pi]$ have?

   A 2 B 4 C 6 D 8

2. Given that $12\sin A - 5\cos B = 9, 12\cos A + 5\sin B = 2$ find the value of $\sin(A - B)$.

   A $-\frac{7}{10}$ B $-\frac{1}{15}$ C $\frac{1}{15}$ D $\frac{7}{10}$

3. Find the trigonometric form of the complex number $\frac{\sqrt{2}}{1-i}$.

   A $\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$ B $2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

   C $\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}$ D $2\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

4. Given that $a$ and $b$ are non-zero vectors. If $|\underline{a} - \underline{b}| = |\underline{a}| = |\underline{b}|$, find the angle between $\underline{a}$ and $\underline{b}$.

   A 30° B 45° C 60° D 75°

5. Given that $x + 1$ is a factor of the polynomial $f(x)$. When $f(x)$ is divided by $(x - 2)(x^2 + 1)$, the remainder is $2x^2 + 3x - 2$. Find the remainder when $f(x)$ is divided by $(x + 1)(x - 2)$.

   A $3x + 3$ B $3x - 3$ C $4x + 4$ D $6x + 6$

6. A group of farmers recorded the daily milk yield (in litres) over five days as 16, 18, 20 22, and 24. Calculate the population standard deviation of milk yield.

   A 4.23 litres B 3.46 litres C 3.16 litres D 2.83 litres

After a rotation of the coordinate axes, the coordinates of the point $(8, -1)$ in the new coordinate system are $(4, n)$. Find the values of $n$.

A $\pm 4$ B $\pm 5$ C $\pm 6$ $\pm 7$

Which of the followings is the polar coordinates of the point with Cartesian coordinates $(-\sqrt{3}, -1)$?

A $\left(2, \frac{7\pi}{6}\right)$ B $\left(2, \frac{5\pi}{6}\right)$ C $\left(2, \frac{\pi}{6}\right)$ D $\left(2, -\frac{\pi}{6}\right)$

9. On which of the following intervals is $f(x) = 8x^4 + 4x^2 - 2$ increasing?

A $\left(-\frac{1}{2}, \infty\right)$ B $(0, \infty)$ C $(-\infty, 0)$ D $\left(-\infty, \frac{1}{2}\right)$

10. A box contains 9 balls, 5 of which are red and the rest are blue. 3 balls are drawn randomly, find the probability that at least one of these 3 balls is blue.

A $\frac{10}{63}$ B $\frac{37}{42}$ C $\frac{4}{27}$ D $\frac{1}{12}$

## Part 2 – Answer ALL questions (Total 70%)

1. Solve the equation: $\sin x - \sqrt{3}\cos x = 1$, where $-\pi < x < \pi$. [4%]

2. Find the range of values of $x$ for which $\left|\frac{x+2}{x}\right| > \frac{1}{2}$. [4%]

3. Given that $\begin{vmatrix} 1 & -1 & 0 \\ -3 & x-2 & 2 \\ 0 & -2 & x+1 \end{vmatrix} = -5$, find the value(s) of $x$. [4%]

4. Given the function $g: \mathbb{R} \to \mathbb{R}, g(x) = x - 2$ and the function $f: [0, \infty) \to (-\infty, 3]$ such that $(g \circ f^{-1})(x) = \frac{x^2-6x+1}{4}$, find the value of $f(9)$. [4%]

5. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and $8y = x^2$ about the $y$-axis through 360°. [4%]

6. (a) Express $\frac{x^4-2x^3+x+2}{x^3-2x^2}$ in partial fractions. [3%]

   (b) Hence, find the value of the definite integral $\int_3^4 \frac{x^4-2x^3+x+2}{x^3-2x^2}\,dx$. [3%]

7. Prove by mathematical induction that for all $n \in \mathbb{N}$,

$$\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \cdots + \frac{2n+1}{n^2 \cdot (n+1)^2} = \frac{n(n+2)}{(n+1)^2}.$$ [6%]

8. Find the parametric equation of the ellipse $9x^2 + 4y^2 - 36x + 8y + 4 = 0$. [3%]
   Hence, or using other methods, find the maximum value of $\alpha + 3\beta$ if $(\alpha, \beta)$ is a point on ellipse $9x^2 + 4y^2 - 36x + 8y + 4 = 0$. [3%]

9. A biologist assumes that there is a linear relationship between the amount of fertilizer supplied to tomato plants and the subsequent yield of tomatoes obtained. Eight tomato plants, of the same variety, were selected at random and treated, weekly, with a solution in which $x$ grams of fertilizer was dissolved in a fixed quantity of water. The yield, $y$ kilograms, of tomatoes was recorded.

| Plant | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| $x$ | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 4.5 |
| $y$ | 3.9 | 4.4 | 5.8 | 6.6 | 7.0 | 7.1 | 7.3 | 7.7 |

(i) Calculate the equation of the regression line of $y$ over $x$. [5%]

(ii) Estimate the yield of a plant treated, weekly, with 3.2 grams of fertilizer. [1%]

10. A particle starts to move from rest along a straight line. After $t$ seconds, the particle moves with a velocity of $v$ $\text{ms}^{-1}$ where $v = t^3 - 10t^2 + 25t$. Find:

(iii) The velocity of the particle at the end of 2nd second. [2%]

(iv) The time when it is at rest again. [2%]

(v) The time when its acceleration is 0. [2%]

11. (a) Solve the differential equation $x(x+1)\frac{dy}{dx} + y = x + 1$. [5%]

(b) Find the intersection point of the straight line $\frac{x-1}{2} = 3 - y = \frac{z+4}{-2}$ and the plane $2x - y + 2z = 0$. [5%]

12. (a) Find $\int x^2 \cos 2x \, dx$. [3%]

(b) Given that the equation of a curve is $y = \frac{1}{(x-1)(x-4)}$.

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [2%]

(ii) Find and determine the nature of the extreme point(s) of the curve. [1%]

(iii) Find the asymptote(s) of the curve. [1%]

(iv) Sketch the graph of the curve. [3%]